

Tue, Apr 21

Reduction Formula

$$\int \sin^n x \cdot dx = \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \cdot dx$$

$$\int \cos^n x \cdot dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \cdot dx$$

$$\int \tan^n x \cdot dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \cdot dx$$

$$\int \cot^n x \cdot dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x \cdot dx$$

①) $\int \cos^2 x \cdot dx$ and $\int \cos^4 x \cdot dx$

②) $\int \cot^2 x \cdot dx$ and $\int \tan^3 x \cdot dx$

$$\int \cot^2 x \cdot dx$$

$$= \frac{-\cot^1 x}{1} - \int \cot^0 x \cdot dx$$

$$= \frac{-\cot^1 x}{1} - \left[-\frac{\cot^1 x}{1} - \int \cot^0 x \cdot dx \right]$$